

CHIBA, Japan

WMO: 476820

Lat: 35.6025N

Lon: 140.1044E

Elev: 6

StdP: 101.25

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.4	1.3	-13.8	1.1	5.0	-12.3	1.3	5.3	13.5	10.3	12.1	9.1	3.3	340	0.469

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 6.0	(c) 32.6	(d) 25.8	(e) 31.6	(f) 25.5	(g) 30.7	(h) 25.2	(i) 26.6	(j) 30.8	(k) 26.2	(l) 30.1	(m) 25.8	(n) 29.5	(o) 5.4	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.5	20.8	29.2	25.1	20.2	28.8	24.6	19.7	28.5	83.5	30.9	81.5	30.2	79.8	29.6	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.0	10.2	8.7	-1.1	35.3	0.7	1.3	-1.7	36.2	-2.1	37.0	-2.5	37.7	-3.0	38.7
			-4.0	27.5	0.9	0.6	-4.7	27.9	-5.2	28.3	-5.7	28.6	-6.3	29.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.5	6.1	6.6	9.7	14.6	19.2	22.2	26.2	27.7	24.2	19.0	13.4	8.6
	DBStd	7.86	2.10	2.62	3.18	3.22	2.66	2.48	2.75	2.37	3.02	2.76	2.82	2.55
	HDD10.0	337	122	102	46	4	0	0	0	0	0	0	4	59
	HDD18.3	1592	379	329	269	116	22	2	0	0	1	24	148	301
	CDD10.0	2713	2	6	36	143	284	365	503	547	426	278	107	16
	CDD18.3	926	0	0	0	4	48	117	245	289	177	44	1	0
	CDH23.3	7392	0	0	0	5	114	496	2283	3076	1289	129	1	0
	CDH26.7	2314	0	0	0	0	7	68	741	1159	328	12	0	0
(14) Wind	WSAvg	3.9	3.8	3.9	4.3	4.5	4.4	3.9	4.3	4.2	3.9	3.5	3.3	3.5
(15) Precipitation	PrecAvg	1467	65	65	108	115	127	151	134	127	205	204	107	59
	PrecMax	1954	155	135	192	209	234	304	314	318	425	605	261	200
	PrecMin	1024	2	6	32	42	59	42	14	10	43	23	9	2
	PrecStd	224	42	34	39	46	45	57	89	84	103	137	60	41
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.4	18.2	20.2	23.9	27.7	30.4	33.9	34.7	32.1	28.2	22.4	19.4
		MCWB	11.2	13.9	13.9	16.6	18.8	22.6	25.4	26.1	25.0	21.2	17.6	15.5
	2%	DB	13.2	15.1	18.5	22.2	25.7	28.5	32.3	33.0	30.8	26.1	21.0	16.4
		MCWB	8.6	11.2	13.7	16.4	18.6	22.4	25.5	26.1	24.8	21.0	16.2	11.6
	5%	DB	11.6	13.0	16.8	20.9	24.6	27.1	31.3	32.1	29.7	24.7	19.7	14.7
		MCWB	6.5	8.6	12.3	15.6	18.3	21.8	25.3	25.8	24.5	20.0	15.1	9.9
	10%	DB	10.4	11.3	15.3	19.7	23.5	26.0	30.2	31.2	28.6	23.4	18.4	13.2
		MCWB	5.3	6.7	11.0	14.8	17.9	21.4	25.0	25.6	24.0	18.9	13.9	8.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.6	14.9	16.2	19.9	22.4	24.8	27.1	27.6	26.4	23.7	19.5	16.4
		MCDB	14.6	18.0	18.2	21.9	24.6	28.8	31.6	32.4	29.9	26.2	21.5	18.9
	2%	WB	9.4	11.8	14.7	18.0	20.8	23.9	26.4	26.9	25.8	22.3	17.5	12.8
		MCDB	12.2	14.5	17.4	20.9	23.5	26.8	30.5	31.1	29.1	25.0	19.8	15.8
	5%	WB	7.6	9.3	13.2	16.7	19.9	23.2	25.9	26.5	25.3	21.1	16.2	10.7
		MCDB	10.8	12.4	16.3	19.7	22.9	26.1	29.9	30.5	28.5	23.6	18.6	13.6
	10%	WB	6.1	7.4	11.5	15.6	19.0	22.3	25.5	26.1	24.8	19.8	14.9	9.3
		MCDB	9.4	10.5	14.7	18.6	22.1	25.1	29.2	29.9	27.9	22.5	17.4	12.3
(35) Mean Daily Temperature Range	5% DB	MDBR	7.1	7.2	7.5	7.3	6.8	5.9	5.8	6.0	5.8	6.2	6.9	7.0
		MCDBR	8.8	9.4	8.8	8.3	8.1	7.2	6.9	6.7	6.5	7.2	8.2	8.8
		MCWBR	6.5	7.4	6.7	5.1	4.2	3.4	2.8	2.5	2.8	4.4	6.0	6.8
	5% WB	MCDBR	8.4	9.1	8.4	7.3	6.9	6.0	6.2	6.3	5.9	6.5	7.5	8.2
		MCWBR	7.2	8.0	7.3	5.5	4.5	3.4	2.8	2.7	3.0	4.5	6.4	7.1
(40) Clear-Sky Solar Irradiance	taub		0.384	0.434	0.503	0.551	0.569	0.596	0.589	0.573	0.510	0.484	0.431	0.387
	taud		2.233	2.086	1.926	1.853	1.868	1.868	1.932	1.964	2.106	2.105	2.166	2.243
	Ebn at Noon		790	787	764	751	744	722	723	726	752	729	728	759
	Edh at Noon		108	140	179	202	202	202	189	179	148	136	115	101
(44) All-Sky Solar Radiation	RadAvg		2.73	3.19	4.02	4.71	5.08	4.50	4.97	5.02	3.85	2.99	2.46	2.38
	RadStd		0.19	0.34	0.26	0.47	0.53	0.42	0.84	0.47	0.34	0.30	0.23	0.19

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+69	N/A
Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page